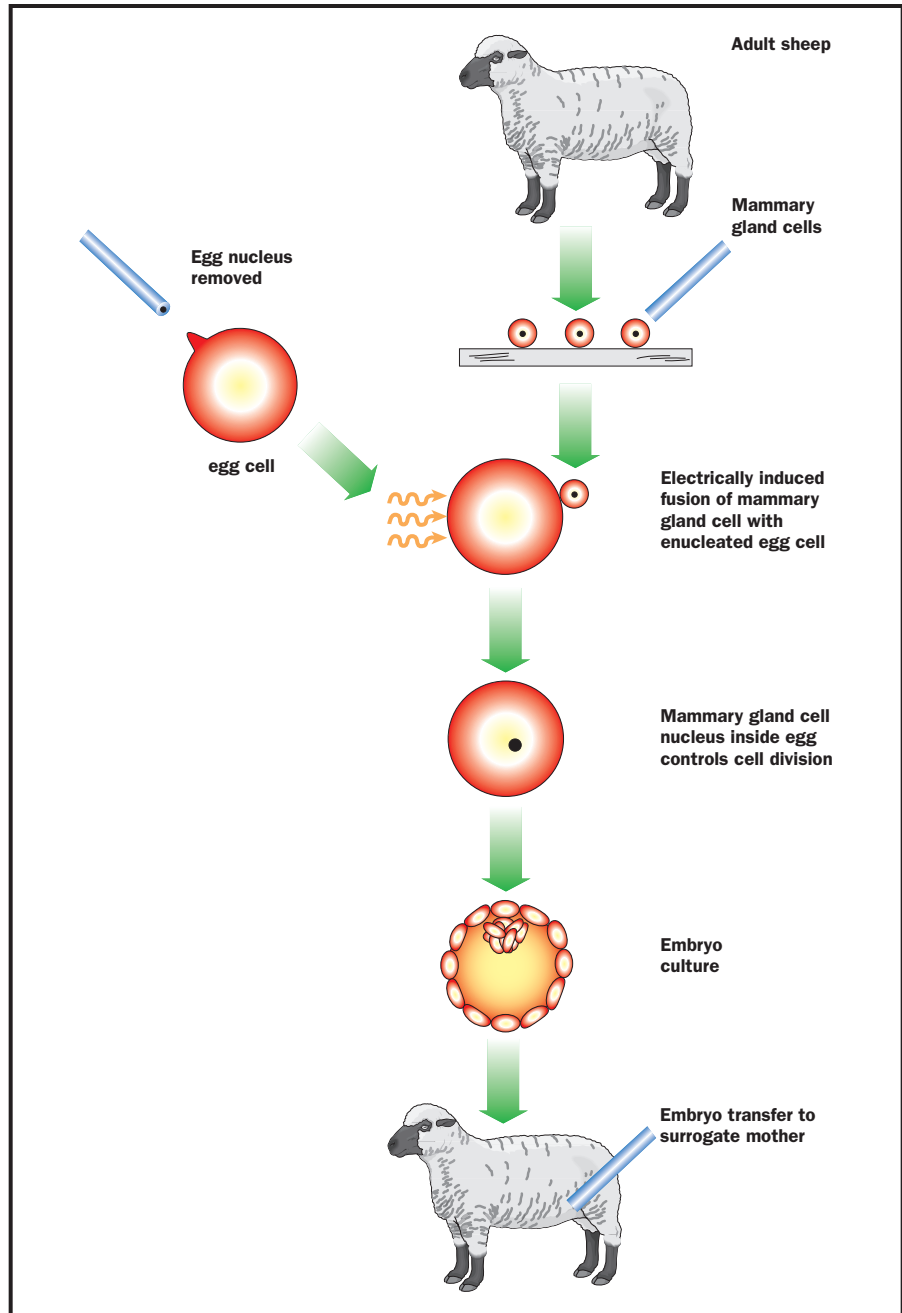


The process used to clone Dolly. Adapted from Gurdon, J. B., 1999.



transplantation, the nucleus of an unfertilized donor egg is either mechanically removed or it is destroyed by ultraviolet light in a process called enucleation. The original nucleus is then replaced by a nucleus containing a full set of genes that has been taken from a body cell of an organism. This procedure eliminates the need for the fertilization of an egg by a sperm.

The most successful nuclear transplants have been achieved after serially transferring donor intestinal nuclei, that is, putting an adult nucleus from an intestinal cell into an egg whose nucleus was destroyed, allowing the egg to divide only a certain number of times, removing nuclei from these cells, and repeating this process several times before allowing the embryo to complete development. Eventually, transplantation of nuclei from albino

embryonic frog cells into enucleated eggs from a dark green female frog led to the production of adult albino frog clones, demonstrating that a properly treated adult nucleus could support the full development of an egg into an adult clone. Later experiments demonstrated that nuclei from cells of other tissues, even **quiescent** cells such as blood cells, could also be used if properly treated. Despite these successes, no adult frog has been cloned when a nucleus from an adult cell was used without serial transfer. Without serial transfer of the nuclei, the animals would only develop to the tadpole stage, and then they would die.

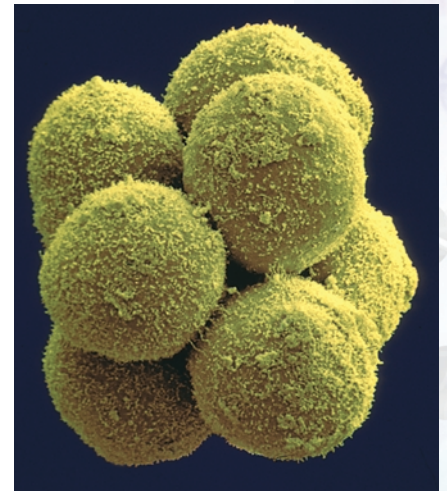
Cloning of Mammals: Dolly

Nuclear transplantation has also been successful in producing mammalian clones, most notably of sheep, cattle, pigs, and mice. The most famous cloned mammal is a sheep named “Dolly,” the first animal to be cloned directly from an adult cell. Experiments leading to the birth of Dolly were done at the Roslin Institute with collaborators at Pharmaceutical Proteins Limited, both in Scotland. This group had earlier produced Megan and Morag, the first mammals to be cloned from cultured cells. These two sheep were produced from embryonic cells, however, not from cells of an adult animal.

Dolly was born in the summer of 1996, the product of a nucleus from the mammary gland of a six-year-old female Finn-Dorsett sheep and an egg from a Scottish Blackface female. Mammary gland cells were grown in a petri dish and were deprived of nutrients so that they would stop dividing, just like an unfertilized egg. Donor eggs were taken from sheep soon after **ovulation**, and nuclei were mechanically removed from them. These enucleated eggs were then fused with the cultured mammary gland cells so that a mammary gland nucleus would be inside an unfertilized egg. Two hundred and seventy-seven such embryos were constructed and temporarily allowed to divide in a petri dish, and then all of them were transferred into the **oviduct** of a temporary surrogate mother. Of the original 247 embryos, only 29 developed further, and these were transferred to 13 hormonally treated surrogate mothers.

Only one surrogate mother became pregnant, and she only had one live lamb, named Dolly. The success rate was very low, but Dolly has been proven to be a true clone: She has all the characteristics of a Finn-Dorsett sheep. Independent scientists used a technique called DNA fingerprinting to show that Dolly’s DNA matched the donor mammary cells but did not match that of other sheep in the Finn-Dorsett flock, nor did her DNA match that of her surrogate mother or the egg donor. Similar results have been obtained by Ryuzo Yanagimachi at the University of Hawaii, who worked with several generations of cloned mice.

In 1997 Polly, a sheep created with a combination of both molecular and organismal cloning techniques, was born. Polly was derived from a fetal sheep cell that had been engineered to contain the human gene that makes coagulation factor IX. Factor IX is missing in people with a disease called hemophilia type B. Polly and two other sheep were engineered to produce factor IX in their milk, thus providing people with hemophilia access to a safer and less expensive source of clotting factor than was previously available. Because Polly was made from more easily cultured and, therefore, more easily engineered embryonic cells, it is thought that this type of cloning



Scanning electron microscope scan of human embryo, day three, at the eight-cell stage.

quiescent nondividing

ovulation release of eggs from the ovaries

oviduct a tube that carries the eggs

technology holds the most promise for the future of pharmaceutical production of proteins that cannot be made in bacteria.

In January 2001, the first cloned member of an endangered species was born. This was a gaur, a wild ox native to India and southeast Asia, which the researchers named Noah. The gaur was chosen by Advanced Cell Technology as a candidate for cloning after the company had successfully cloned domestic cattle, which are related to the gaur species.

The embryo from which Noah developed was created from the nuclei of frozen skin cells that had been taken from an adult male gaur that had died eight years earlier. Skin cell nuclei were fused with enucleated domestic cow eggs to produce forty embryos. One of these forty was carried to full term in a surrogate cow mother. Unfortunately, Noah died of an infection two days after his birth (the infection is thought to be unrelated to his origin as a cloned animal). Despite Noah's death, it is likely that cloning will eventually be used to aid the conservation of endangered species. In the future, scientists may attempt to clone a recently extinct species, should intact DNA for an extinct species be obtained.

Problems with Cloning

In general, the success rate of mammalian cloning is low, with less than 0.1 to 2.0 percent of transplanted nuclei yielding a live birth. The vast majority of transplants fail to divide or to develop normally, indicating there is much we still do not understand about reprogramming an adult nucleus to support embryonic development. One thing that is clear, however, is that having both the donor cell and host egg cell in a nondividing state is essential for success.

What might be both the most vexing and most interesting problem with cloning is related to aging. Chromosomes “show their age” by a shortening in their tips, or **telomeres**, a process that occurs every time the cell they are in divides. This telomere shortening occurs in all cells except eggs, sperm, and most cancer cells, and shortened telomeres are correlated with the aging of organisms. Since the nuclear DNA in most cloned animals is taken from an adult, the chromosomes of cloned animals are expected to have shorter telomeres than animals of the same birth age that are produced by sexual reproduction, causing researchers to wonder whether cloned animals will age prematurely. Shorter telomeres have been found in Dolly and other cloned sheep, but telomeres are reported not to be shorter in cloned mice or cattle. Underlying reasons for the different results may include differences between cell types or species used.

The Myth of the Perfect Clone

Cloned animals are not 100 percent identical to their “parents.” Whenever nuclear transplantation is used to produce cloned organisms, the offspring display some differences from the organism that donated the nuclei. The egg donor contributes mitochondria, the energy producers of eukaryotic cells, and these mitochondria have their own small amount of DNA-containing genes used for energy metabolism. Since mitochondria are inherited only with egg cytoplasm, they will not match the mitochondria of the animal from which the nucleus was taken. In addition, maternally derived gene products, both mRNA (messenger RNA) and protein, which serve to

telomeres chromosome tips

begin embryonic development, will differ from that of the nuclear donor, as will the uterine environment and the external environment. Thus, for example, clones produced by nuclear transplantation will be significantly less identical than will clones produced by twinning. SEE ALSO CLONING: ETHICAL ISSUES; CLONING GENES; CONSERVATION BIOLOGY: GENETIC APPROACHES; HEMOPHILIA; MITOCHONDRIAL GENOME; REPRODUCTIVE TECHNOLOGY; TELOMERE; TRANSGENIC ANIMALS; TWINS.

Elizabeth A. De Stasio

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College Professor

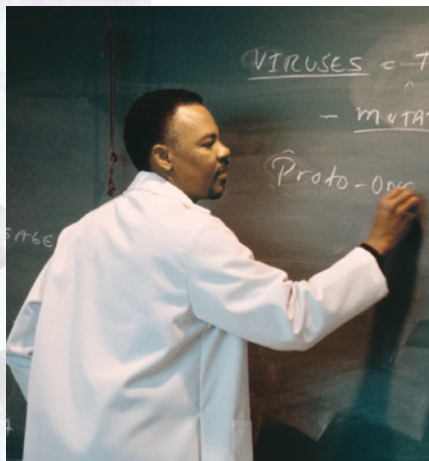
Professors of genetics at colleges and universities teach, perform research, and handle administrative responsibilities. Professors may teach at the undergraduate or graduate levels, or both. Undergraduate teaching involves class lectures, small group seminars, and hands-on laboratory sessions. Professors evaluate students based on their performances on examinations, essays, and laboratory work, and may also work individually with students to advise them about their college careers or to mentor them in independent laboratory studies. Graduate teaching provides advanced instruction in the field of genetics, usually in smaller classes. Professors also act as mentors to graduate students, providing a supportive environment for conducting research.

Research is an integral component of professorship. Professors apply for grants to fund genetic experiments, perform original research, analyze the results, and submit their findings for publication. They must keep current with the published results of other scientists in the genetics field by reading journals and books, attending the conferences of professional societies, and interacting with other researchers. In addition, professors must fulfill administrative responsibilities, including participation on departmental and faculty committees that consider issues such as courses of study for students (curricula), budgets, hiring decisions, and allocating resources.

Most genetics professors at four-year universities hold a doctorate degree (Ph.D.) in a specialized area of genetics or molecular biology. Earning a Ph.D. first requires completion of an undergraduate bachelor's degree. The student must then complete graduate school, typically consisting of about three years of advanced coursework and two to three years of original, independent laboratory research. The results of this research are written up in an extensive report, called a dissertation.

The career path of a college professor can be described as a rise through four levels: instructor, assistant professor, associate professor, and full professor. Instructors are either completing or have already earned their Ph.D., and are beginning their teaching careers. They usually spend about nine to twelve hours a week teaching. The *Occupational Outlook Handbook*, published





Genetics professors are often specialists in a specific area of molecular biology or genetics and are expected to juggle the roles of educator and researcher. Here, a biology lecture is given by a professor at Morehouse College in Atlanta, Georgia.

by the U.S. Department of Labor, lists the average instructor's salary as \$33,400. After instructors have taught for several years, their teaching, research, and publications are reviewed by their academic institution. If found satisfactory, they are eligible for promotion to the position of assistant professor. Assistant professors also teach nine to twelve hours weekly, but are more likely to lecture in large undergraduate courses. They are expected to conduct research projects and publish their work. Assistant professors earn approximately \$43,800.

With continuing success in research and teaching responsibilities, a professor can obtain the position of associate professor. These faculty members spend fewer hours on undergraduate teaching (about six to nine hours a week), and are likely to lead graduate classes and advise graduate students on their dissertation projects. They can expect to make about \$53,200 yearly. Promotion to full professorship is based on the quality of one's research and reputation with the field. Teaching is less emphasized, usually occupying only three to six hours per week. Professors take an active role in the research projects and dissertations of doctoral candidates. Further advancement opportunities include positions in administration such as department chair, dean of students, or college president.

College professors may work at public or private institutions. They generally teach for nine months of the year, allowing them to work in other environments as well. Instructors may teach additional classes, act as consultants to private, governmental, or nonprofit organizations, or author publications in their field of expertise. The ability to make one's own schedule, conduct original research, teach and mentor students, take paid leaves of absence, and have access to campus facilities makes professorship an attractive and competitive choice for motivated individuals. SEE ALSO EDUCATOR; GENETICIST.

Regina M. Carney

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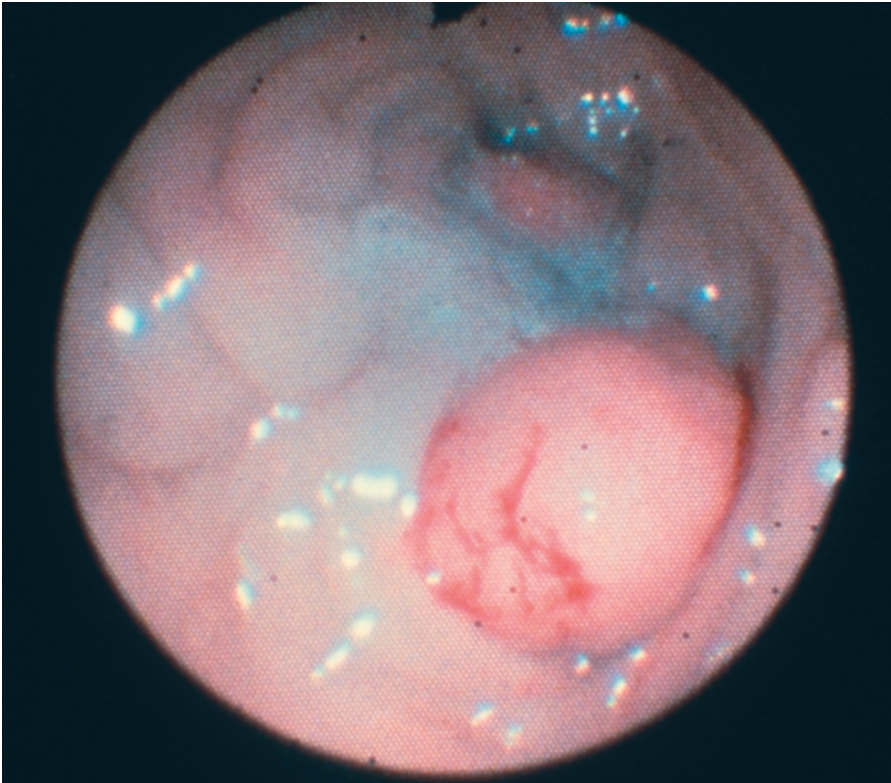
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Colon Cancer

Colon cancer is the second leading cause of cancer death in the United States, occurring in approximately 5 percent of the population and resulting in roughly 55,000 deaths annually. New cases of colorectal cancer are diagnosed in approximately 90 per 100,000 people annually. The majority of cases occur in individuals older than age fifty. Of those who suffer from colorectal **malignancy**, an estimated 40 percent will die from the disease.

Colon cancer-related health-care costs, consisting of outpatient visits, hospitalizations, hospice and home health care, medications, and physician services, exceed \$5 billion per year. This figure does not include the indirect costs of wages lost and reduced productivity.

malignancy cancerous tissue



adenoma a tumor (cell mass) of gland cells

The colon, also known as the large intestine, is the final portion of the digestive tract and consists of a tube (called the lumen) lined by specialized cells called colonic epithelial cells. These cells are constantly reproducing in a regulated manner, but when the growth and division of colonic epithelial cells becomes unregulated, colon cancer may result.

Major Genes Involved

Colorectal cancer may develop in an individual who has a strong inherited risk. This occurs, for example, in patients with familial adenomatous

alleles particular forms of genes

apoptosis programmed cell death

endoscopic describes procedure in which a tool is used to see within the body

colectomy colon removal

autosomal describes a chromosome other than the X and Y sex-determining chromosomes

polyposis (FAP) and hereditary nonpolyposis colon cancer (HNPCC). These patients will usually have a specific genetic alteration.

APC. The Adenomatous Polyposis Coli (*APC*) gene is found on chromosome 5 and is a tumor suppressor gene. Both **alleles** of the gene must be inactivated for tumor growth to occur. In the normal cell, the *APC* gene plays a role in regulating the cycle of cellular division and replication, as well as in cell-to-cell communication, thereby suppressing tumor development. Mutations of *APC* result in a loss of gene function, thus allowing unregulated cellular proliferation. *APC* mutations are found in the majority of common colon polyps and cancers and in patients with FAP, and they may be one of the earliest genetic alterations in the adenoma-carcinoma sequence.

K-ras. The *K-ras* gene plays an active role in cellular signaling and promoting cell growth. The normal gene exists in both an active and inactive form. However, in the abnormal state, the active form predominates and results in a continually growth-stimulated state.

p53. The normal *p53* gene is responsible for regulating cells with damaged DNA by directing abnormal cells either to halt the cycle of cell division or to die as the result of a process called **apoptosis**. Like *APC*, the *p53* gene is a tumor suppressor. With the *p53* mutation, the gene no longer functions, and this permits the uninhibited proliferation of cells that may have damaged DNA. *p53* mutations are seen in more than half of colorectal cancers.

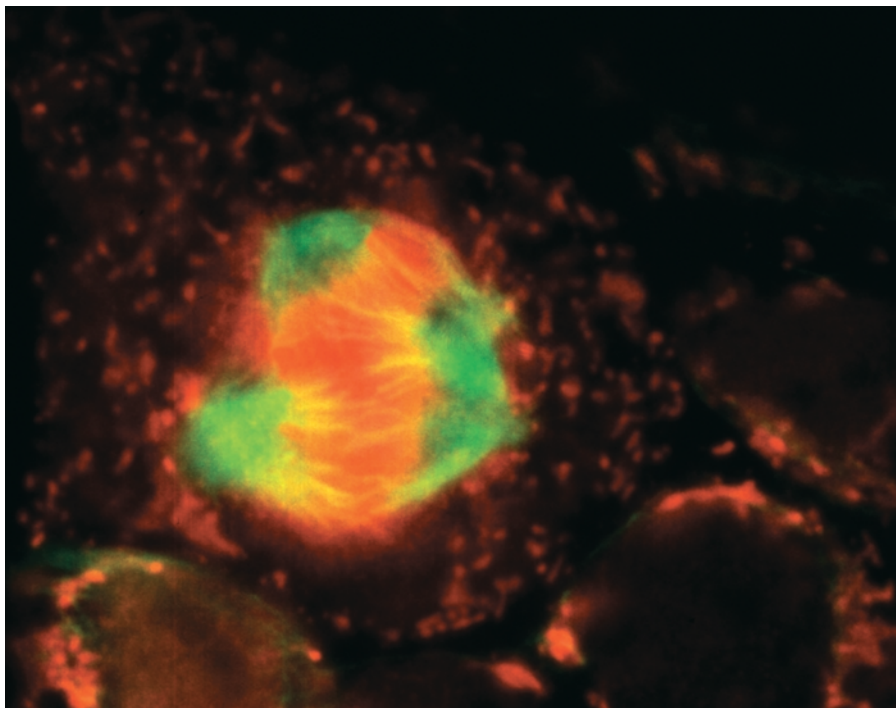
Familial Adenomatous Polyposis (FAP)

FAP is characterized by the presence of hundreds or thousands of colonic polyps, and it accounts for less than 1 percent of colon cancers. Affected individuals are at increased risk of colorectal polyps and malignancy, and they usually develop polyps by age thirty-five and cancer by age forty. Because **endoscopic** removal of all the polyps is impossible, patients are usually advised to consider **colectomy** at a relatively young age. In addition to occurring in the colon, polyps occur in the upper digestive tract, and a variety of tumors may develop outside the gastrointestinal tract. Because mutations in the *APC* gene can be identified in most cases, genetic testing is now available for affected families.

FAP is inherited as an **autosomal** dominant disorder with 95 percent penetrance, meaning 95 percent of those who inherit one mutated *APC* gene will develop FAP. At the cellular level, the *APC* mutation is actually recessive: As long as there is one copy that is not mutated, the cell cycle will remain controlled.

The apparent paradox of a recessive gene causing a dominant disorder is resolved when we consider how a gene defect predisposes a person to cancer. Of the many millions of cells lining the colon, it is highly likely that some will undergo spontaneous mutation in one of the two copies of the *APC* gene.

For a person not affected by the *APC* gene, a spontaneous mutation of one allele will not lead to cancer, because the other gene copy remains intact. However, for a person affected who inherits one mutated copy of the *APC* gene, each of the cells lining the colon begins with one bad gene copy. Any mutation to the remaining good copy will cause the cell to lose control of



Digital image of multipolar spindles (green) separating the chromosomes (red) of a cancerous colonic epithelial cell in a late stage of its division.

its cell cycle and begin the process of polyp development. Given the number of cells involved, it is almost inevitable that this will occur in some of them. Hence FAP is a dominant condition.

Hereditary Nonpolyposis Colon Cancer

HNPCC is an autosomal dominant disorder that may be responsible for up to 5 percent of colon cancers. The genetic mutation leading to the abnormality is the mutation of DNA mismatch repair genes. Individuals with this mutation have up to an 80 percent chance of developing colon cancer. At least five genes are involved in this syndrome.

Malignancy in patients with HNPCC occurs at a younger age than in the general population, is more often located in the proximal colon (the portion nearest the small intestine), and may be associated with multiple tumors. HNPCC also carries an increased risk of tumors of the **endometrium**, ovaries, stomach, small intestine, bile ducts, bladder, renal pelvis, and ureters. Genetic testing is useful for HNPCC families.

Genetic Testing

Genetic testing may benefit patients and families affected by an inherited colon cancer syndrome. The genetic mutations in FAP and HNPCC can often be characterized. If a family member with colon cancer has an identified genetic abnormality, other family members can be tested to see if they have the same abnormality.

If the mutated gene is not found, the abnormal gene was not inherited and the family member is not at increased risk of developing cancer. Because screening (i.e., colon examination) is more frequently performed for those with FAP and HNPCC than others, genetic testing can be useful for determining who would benefit from intense surveillance.

endometrium uterine lining



Other Risk Factors

In addition to the well-described genetic syndromes of FAP and HNPCC, other factors that place an individual at increased risk include a personal or family history of colon cancer and the presence of inflammatory bowel disease (e.g., ulcerative colitis and Crohn's disease). Population studies support an association between the development of colon cancer and a high-fat, low-fiber diet, although a cause-and-effect relationship has not been proved.

Prevention

Inhibiting the development of polyps and cancers, finding and removing premalignant polyps, and testing individuals at high risk may reduce colon cancer-related morbidity and mortality. Colon cancer occurs less commonly in individuals whose diets are high in calcium and folate, who take multivitamins, and who maintain high-fiber and low-fat diets.

Non-steroidal anti-inflammatory medications like aspirin may reduce the numbers of polyps, particularly in families with FAP. Colonoscopy can identify polyps that may be premalignant and can facilitate polyp removal. It is recommended that all individuals have a colonoscopy at age fifty. High-risk patients, such as those with inflammatory bowel disease, FAP, or HNPCC, should have screening initiated at an earlier age and repeat exams at shorter time intervals. SEE ALSO APOPTOSIS; BREAST CANCER; CANCER; CARCINOGENS; CELL CYCLE; DNA REPAIR; GENETIC TESTING; MUTATION; ONCOGENES; TUMOR SUPPRESSOR GENES.

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Color Vision

nanometer 10^{-9} meters;
one billionth of a meter

Sight is a complex process that results when visible light, a narrow band of the electromagnetic spectrum between 400 and 700 **nanometers** (nm), is converted into signals that can be interpreted by the brain. This process involves special light-sensitive cells called photoreceptors that are located in the retina, a thin structure that lines the inside of the eye. These cells capture packets of light, called photons, and transform their energy into signals that are transported from the eye to the occipital cortex, the portion of the brain that allows us to interpret these signals as sight.

Normal human color vision is trichromatic (based on the perception of three primary colors) and requires three types of photoreceptor cells, called cones, each of which contains a different photopigment. Each photopigment

CLASSIFICATION AND INCIDENCE OF COLOR VISION DEFECTS

Color Vision	Inheritance	Incidence in Male Population (Percent)
I. Hereditary		
<i>Trichromats</i>		
1. Normal		92.0
2. Deuteranomalous	XR	5.0
3. Protanomalous	XR	1.0
4. Tritanomalous	AD	0.0001
<i>Dichromats</i>		
1. Deuteranopes	XR	1.0
2. Protanopes	XR	1.0
3. Tritanopes	AD	0.001
<i>Monochromats (achromats)</i>		
1. Typical (rod monochromats)	AR	0.0001
2. Atypical (cone monochromats)	XR	Unknown
II. Acquired		
1. Tritan (blue-yellow)		
2. Protan-deutan (red-green)		

Table 1. Adapted from American Academy of Ophthalmology, 1995.

absorbs particular wavelengths of light in the short (blue, 440-nm), middle (green, 545-nm), or long (red, 560-nm) wavelength region of the visible spectrum. About 7 percent of all cones are blue-sensitive, 37 percent are green-sensitive, and 56 percent are red-sensitive. These cones are the basic mediators of color vision. If one or more of their pigments is missing, color blindness results. Rod cells, unlike cones, detect light intensity but not color. The photopigment in rod cells is called rhodopsin.

The spectral sensitivity of the cone photopigments is intimately related to the structure of the pigment molecules. These are concentrated in the photoreceptor outer segment, the portion of the cell containing the photo-transduction machinery. Each pigment molecule consists of an opsin protein and a chromophore (11-cis-retinal), which is a derivative of vitamin A. Photon absorption by the pigment molecules causes a change in the shape of the chromophore, which initiates the processes that lead to vision.

The different opsins of the cone photopigments and of the rod photopigment are encoded by four separate genes, the *BCP* (blue cone pigment), *GCP* (green cone pigment), *RCP* (red cone pigment), and *RHO* (rhodopsin) genes. The genes encoding the blue cone and rod pigments reside on the long arms (called the q arms) of chromosome 7 and chromosome 3, respectively. The genes encoding the red and green cone pigments reside on the q arm of the X chromosome.

Color vision defects may be divided into two groups, hereditary and acquired. Hereditary color vision defects are almost always “red-green” and affect 8 percent of males and 0.5 percent of females. Acquired defects are more often “blue-yellow,” and affect males and females equally. Hereditary defects are typically bilateral (affecting both eyes), while acquired defects may affect one eye only and are often asymmetric. Hereditary color vision defects tend to remain stable throughout life and are usually not associated with other retinal or optic nerve **pathology**. Acquired defects, however, may have a more variable course and are frequently associated with observable

pathology disease
process

Table 2. Adapted from American Academy of Ophthalmology, 1995.

COLOR VISION TESTS

Test	Sensitivity/ Quantification	Ease of Administration
AO-HRR	Will miss very mild R-G/good classification	Excellent for all ages
Farnsworth-Munsell 100 hue	Extremely sensitive/classify by error scoring	Tedious to administer
Ishihara	Extremely sensitive/nil	Difficult for pre-school children and low-IQ patients
Farnsworth's Panel D-15	Will only detect severe anomalous trichromats and dichromats/good classification	Easy to administer
Nagel's anomaloscope	Very sensitive/classify by anomaly (R-G) quotient	Good
Sloan's achromatopsia test	Grossly sensitive/very incomplete achromatopsia pass	Easy to administer

NOTE: All tests, with exception of Nagel's anomaloscope, are to be administered under an illuminant C source such as provided by Macbeth easel lamp.

ocular pathology. A common cause of acquired color-vision loss is optic nerve disease, such as optic neuritis.

Inherited color blindness usually results from the loss of one of the photopigments and reduces color vision to two dimensions, or dichromacy. Other less common conditions reduce color vision to one dimension (monochromacy), or may completely extinguish it (achromacy). Vision in this last circumstance is purely dependent on the rods, which function primarily in dim conditions and do not contribute to color vision.

The most common forms of hereditary color blindness are protanopia/anomaly and deutanopia/anomaly, both of which are caused by defects in the red (L) and green (M) cones. Also known as red-green color vision deficiencies, they typically demonstrate an X-linked recessive pedigree pattern. The incidence of X-linked color-vision defects varies between human populations of different racial origin, with some of the highest rates appearing in Europeans and some populations in India.

The incidence of these common forms of color blindness is much lower in females than in males because the defects are inherited as X-linked recessive traits. Males, who have only one X chromosome, are hemizygous (meaning that they have only one gene present for the trait) and they will always manifest a color vision deficiency if they inherit an abnormal gene from their mother. Females, on the other hand, have two X chromosomes, one inherited from each parent, so they will not usually show a complete manifestation of the typical color defect unless they are homozygous, though a partial manifestation of color blindness may be present in heterozygotic carriers. A variety of special tests are used to screen for these red-green color-vision defects. SEE ALSO INHERITANCE PATTERNS; MOSAICISM; SIGNAL TRANSDUCTION; X CHROMOSOME.

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Combinatorial Chemistry

Combinatorial chemistry is a technology for creating a multitude of different compounds by reacting different combinations of interchangeable chemical "building blocks." The compounds are then screened for their ability to carry out a specified function, most commonly to act as drugs to treat a disease. Combinatorial chemistry allows the rapid synthesis and testing of many related compounds, greatly speeding the pace of drug discovery. Automated synthesis and screening systems are key to this approach.

The Combinatorial Approach

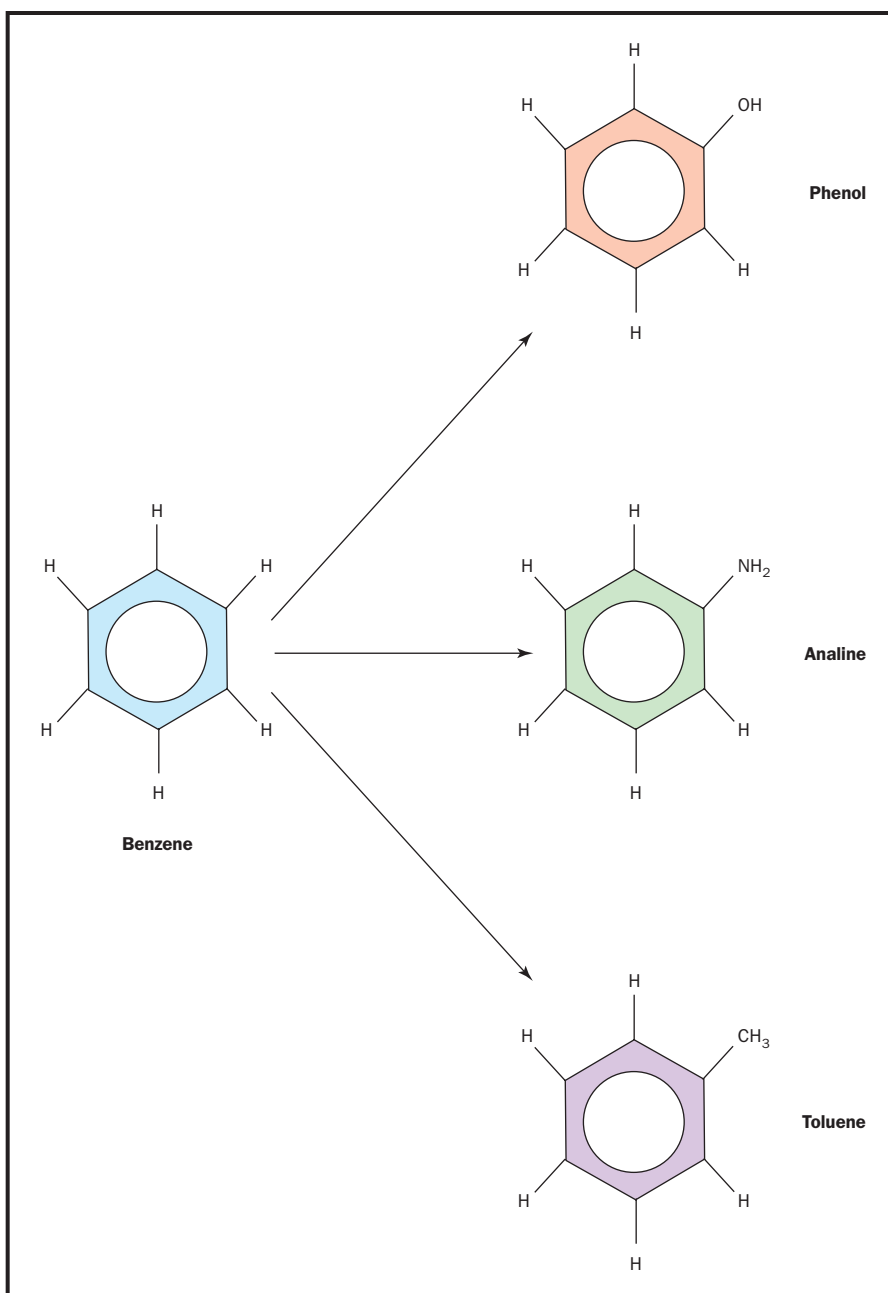
There are two general approaches for finding the correct answer to a question (besides asking someone who knows). One way is to learn everything relevant to the topic and then to use your knowledge to arrive logically at the answer. Scientists, and most other people, almost always use this method. A second approach is to keep guessing until you've guessed right! This seems like a foolhardy strategy, and usually is. What if it took a million guesses before you stumbled upon the right answer? But what if you could make a million or a billion guesses all at once? Through combinatorial chemistry, scientists can make and test millions, billions, or even quadrillions (10^{15}) of guesses about which chemical compound might have a desirable function, such as the ability to bind to a specific molecule, or to serve as a drug.

Many chemicals are pieced together through combinations of smaller building blocks. For example, benzene is a chemical consisting of six carbon atoms connected in an aromatic ring structure, with a hydrogen atom bound to each carbon. Substituting one of the hydrogens with a hydroxyl ($-\text{OH}$) group forms the chemical phenol. Substituting a methyl ($-\text{CH}_3$) group instead forms toluene, and substituting an amino ($-\text{NH}_2$) group forms aniline. Because of their different "functional groups," or side groups, all of these compounds have very different physical and chemical properties. More variations can be synthesized by substituting additional side groups with more than one of the hydrogens. By substituting one of just these three groups (or by not adding any groups) for any of the six hydrogens in a benzene ring, there are 4^6 , or 4,096, possible combinations (the number of different compounds is much smaller, because benzene is symmetrical, and many of the combinations represent equivalent structures).

Side groups can also be placed onto other side groups. For example, a single chlorine atom can substitute for one of the hydrogens of the methyl group in toluene to form benzyl chloride. By using a moderately sized collection of side groups, placing them onto a "scaffold" molecule that is more complex than



The substitution of side groups on simple molecules can create many new molecules.



synthesize create

benzene (such as cholesterol, which has three six-carbon rings and a five-carbon ring), and by using additional levels of side groups, combinatorial chemists can **synthesize** vast numbers of distinct but related compounds.

Although the utility of combinatorial chemistry was not fully appreciated by scientists until the 1980s, nature uses this strategy over and over. Genes, after all, are composed of different combinations of only four different nucleotides, and just twenty different amino acids form the building blocks of all proteins. In the immune systems of mammals, B lymphocytes use an elaborate scheme for mutating and combining different segments of antibody genes to generate a diverse pool of antibody molecules that can recognize and bind a wide array of alien molecules that enter the body with a pathogen infection.

Drug Targets

Combinatorial chemistry is most often used to synthesize “small molecules,” in contrast to **macromolecules** such as DNA, RNA, proteins, and polysaccharides, which are polymers containing long chains of monomer subunits. Because of their enormous size, macromolecules cannot easily enter cells, which is an important requirement for compounds intended for use as drugs.

In many cases the combinatorial chemist is looking for a compound that will bind tightly and specifically to a cellular molecule, such as the catalytic, or “active,” site on the inside of an **enzyme**. Small molecules can fit into the holes and crevasses leading to the active site. By binding the enzyme, the synthetic compound may prevent it from binding to its natural substrate or from carrying out its catalytic reaction. Defective enzymes that resist normal cellular restraints on their activities are responsible for many diseases, including certain cancers. Chemical inhibitors of such rogue enzymes hold promise as powerful drugs. Alternatively, binding of a small molecule to an enzyme could enhance the enzyme’s normal activity. Such molecules have potential as drugs for diseases caused by insufficient activity of a crucial enzyme.

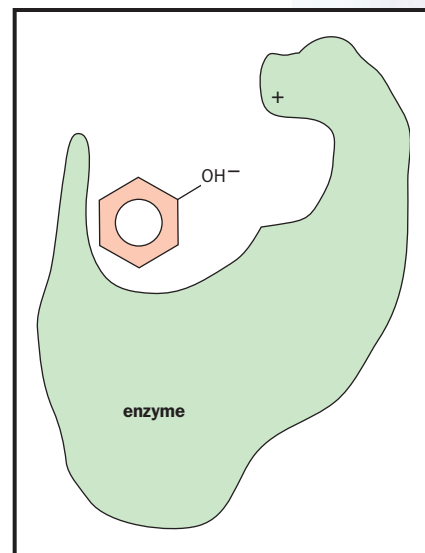
For two molecules to bind to one another, they must have a proper fit, like a key in a lock. The fit depends on the shapes of the two molecules as well as on the chemical interactions between them. For example, two positively charged side groups will repel each other, but negative and positive groups can attract. Not surprisingly, a synthetic compound that binds a particular molecule often has chemical properties and a shape mimicking the natural ligand for the molecule. Such compounds are termed analogues.

When the drug target is known, its structure can be used as a **template** to create analogues with complementary shapes. Alternatively, if an analogue is already in hand that binds the target but has undesirable properties (such as weak binding, poor solubility, or serious side effects), this structure can be used as a starting point. Even without such clues, the speed and automation of the combinatorial approach makes it feasible to randomly synthesize and test millions of compounds.

High-Throughput Screening

A library of a billion or more different molecules is only useful if the molecules can be quickly and economically screened for the desired function. “High-throughput” techniques have been developed that automate most of the steps required to combine the molecules with their targets and evaluate the extent of any reaction.

Typically, the molecules are arrayed on a solid surface and the target is added. Unbound target is washed away. Fluorescent tags are often added to the target, to allow easy (and automated) visualization of the results. Robotic systems controlled by computers can react and evaluate billions of separate compounds in the time it would take a human to screen a dozen. One such approach is used in DNA microarrays, in which thousands of genes from a DNA library are attached to a solid base. These are reacted with messenger RNAs from a cell, and the results are visualized fluorescently.



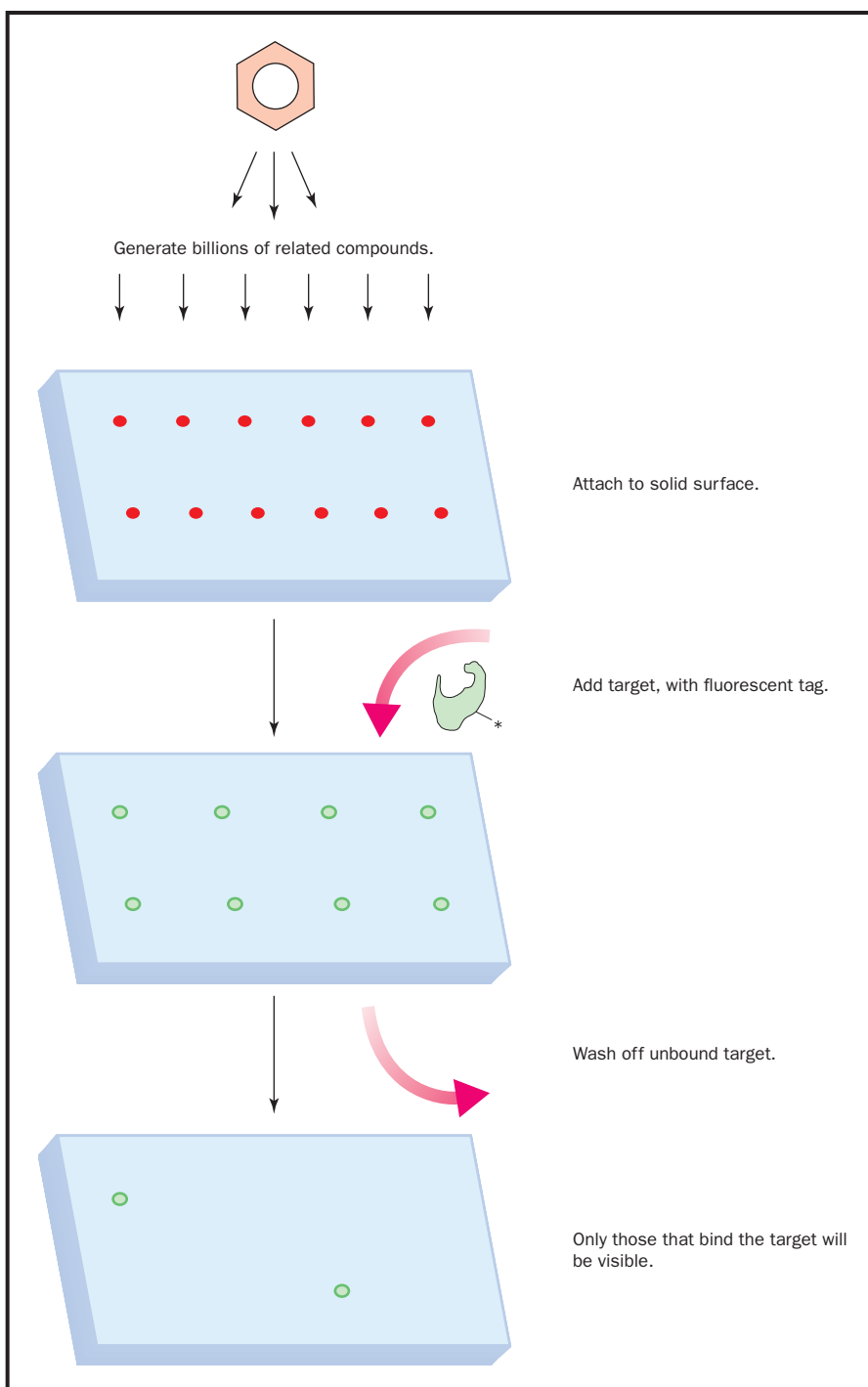
Interaction of small molecules with target enzymes depends on shape and charge.

macromolecules large molecules such as proteins, carbohydrates, and nucleic acids

enzyme a protein that controls a reaction in a cell

template a master copy

A high-throughput screening assay can quickly assess the target-binding ability of millions of compounds.



SELEX

In addition to its use in drug development, combinatorial chemistry can be applied to other areas of biomedical research, such as the design of molecules for diagnosing medical conditions. Compounds for these applications can be larger than pharmaceutical compounds, and do not have to be designed to enter the body. Using a novel combinatorial chemistry method called *in vitro* selection or SELEX (systematic evolution of ligands by exponential enrichment), a short DNA or RNA molecule (termed an oligonu-

cleotide) with a desired property, such as the ability to specifically recognize and bind a molecule associated with a particular disease, can be selected in a single experiment from a library containing approximately 10^{15} different compounds. First, a library of oligonucleotides is created in a machine called an oligonucleotide synthesizer. This apparatus can make oligonucleotides with either a defined or random sequence.

Oligonucleotides for SELEX are designed to have a central region containing random sequence and outer, flanking regions with defined sequences. These defined sequences will be used as primer-binding sites for the polymerase chain reaction (**PCR**). The oligonucleotide library is prepared as a mixture, usually containing about 10^{14} to 10^{15} different sequences. These specialized oligonucleotides, termed aptamers, are then exposed to target ligand molecules, which are typically attached to a solid support, such as a filter membrane. The unbound aptamers are then washed away, leaving only the rare aptamers that can bind the ligand adhering to the filter. These aptamers can then be recovered from the filter by washing it with a solution that disrupts the binding.

These binding candidate aptamers represent a minuscule fraction of the original library. Some may bind the target ligand tightly, but others may bind weakly. Since all the aptamers have defined primer-binding sites on the ends, this much-reduced population can now be amplified exponentially by PCR. After amplification, the aptamers can be subjected to another round of ligand binding, now using more stringent washing conditions, in which only the tightest-binding molecules will stay bound. These high-affinity binders can be recovered again subjected to still more cycles of PCR amplification, binding, washing, and recovery, until the population of aptamers consists exclusively of very tightly binding molecules.

For some applications, these molecules are useful directly. They can also be studied to design non-DNA molecules that have similar shapes but that will have more potential as drugs. **SEE ALSO** DNA LIBRARIES; DNA MICROARRAYS; HIGH-THROUGHPUT SCREENING.

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Complex Traits

Complex traits are those that are influenced by more than one factor. The factors can be genetic or environmental. This is in contrast to simple genetic traits, whose variations are controlled by variations in single genes. Examples of simple traits include Huntington's disease and cystic fibrosis. Each of these traits is caused by a **mutation** in a single gene that alters or destroys the function of that gene. There are several thousand disorders caused by single genes, but these are almost always quite rare in the population, often occurring in less than one in five thousand individuals.

PCR polymerase chain reaction, used to amplify DNA

mutation change in DNA sequence

Almost any trait that is not simple is considered complex. If there are just a few genes that affect a trait, it may be called oligogenic. If there are many different genes that affect a trait, it may be called polygenic. If other, nongenetic factors are involved, it may be called multifactorial.

Diseases inherited as complex traits are often much more common in the population and include heart disease, Alzheimer's disease, and diabetes. There are many factors that can affect a complex disease. Perhaps most commonly, these traits have multiple genes, where variations in those genes can influence the risk of developing the disease.

How Genes Are Involved in Complex Disease

Because there are multiple factors involved in complex traits, such traits are difficult for scientists to study and even more difficult to understand. This is because the different factors may not all act equally or independently on the trait.

For example, if there are three genes involved in a trait, the simplest hypothesis is that each gene will contribute about one-third of the genetic effect on the trait. If the effect of the variations in the genes can be added together, then this is called an additive effect. However, that is not the only way multiple genes can have an effect. Rather than being added together, the effects of variations in the genes may have to be multiplied together. If this is the case, this is called a multiplicative effect. Both parts of figure 1 show examples of additive and multiplicative effects.

Both additive and multiplicative effects imply that a variation in each gene has an effect and that the overall effect gets larger when additional genes are involved. However, in some cases two or more gene variants may need to occur together before any effect is seen. An example of this is shown in Figure 2. If multiple genetic variants must occur together, then this is called an epistatic interaction. In epistasis, a particular form of one gene must be present for the effect of a second to be felt.

The overall effect of genes on a trait can be even more complicated, because genes may act in combinations of additive, multiplicative, and epista-

		Variation in Gene 1		
		GG	GT	TT
Variation in Gene 2	GG	0	2	4
	GT	2	4	6
	TT	4	6	8

Figure 1A. In this model two-gene system, the risk of developing a disease varies between 0 and 8. It is determined by adding risk (2) for each T a person has at both Gene 1 and 2.

Variation in Gene 2		Variation in Gene 1		
		GG	GT	TT
GG		1	3	9
GT		3	9	27
TT		9	27	81

Figure 1B. Here, the risk of developing a disease varies between 0 and 81. It is determined by multiplying the risk (3) for each T a person has at both Gene 1 and Gene 2.

tic ways. For example, if there are six different genes whose variations can influence the risk of developing the disease, Genes 1, 2, and 3 may act additively, Genes 4 and 5 may act multiplicatively, and Gene 6 may act in an epistatic manner with Gene 2. Another level of complexity can occur because different variations within the same gene may act differently. For example one variation (**allele**) in Gene 1 may act additively with Gene 2, whereas another may act multiplicatively with Gene 2.

In simple traits, the variations within the gene usually create major changes in the way the gene's product (the protein it codes for) acts. In most cases, these changes (mutations) are considered causative, because having them is enough to cause the disease. In other words, having the mutation is sufficient to get the disease. These are called causative genes. For instance, in Huntington's disease, the presence of the expanded form of the *huntingtin* gene is sufficient to cause the disease.

In contrast, in complex traits, the variations in any one of the genes are not usually enough to cause the trait. These variations may simply increase (or decrease) the probability of developing the disease. Thus these genes, and the variations within them, are often called susceptibility genes and susceptibility alleles. It is a particular combination of susceptibility alleles across multiple genes, and possibly including environmental factors, that causes a complex disease.

How the Environment Is Involved in Complex Disease

Genes are not the only things that can affect a complex trait. Often environmental factors can also be involved. The type of environmental factors can be very different for different traits. One obvious example of this is lung cancer. Smoking cigarettes greatly increases the risk of developing lung cancer. Smoking also seems to have an effect on other diseases, including some eye diseases (such as age-related macular degeneration). However, not every chronic smoker will develop lung cancer or eye disease: The presence of particular alleles of susceptibility genes is also a risk factor, as discussed below.

allele a particular form of a gene



Figure 2. Here, the risk of developing a disease is either 0 or 9. It is determined by having at least one T at both Gene 1 and Gene 2.

Variation in Gene 2	Variation in Gene 1		
	GG	GT	TT
GG	1	1	1
GT	1	9	9
TT	1	9	9

Other environmental factors, which may be more difficult to identify directly, can still be identified by measuring other risk factors, such as gender, age, occupation, level of education, use of alcohol, and other measures of lifestyle. For example, the effect of estrogen may not be measured directly, but could be identified by finding an effect of gender on the risk of developing the disease. The strength of an environmental effect may vary by how much exposure a person has to that effect. For example, the number of years someone has smoked or how many cigarettes per day someone smokes may change the effect of exposure to smoke.

Genes and Environmental Factors May Interact

Just as with the genes themselves, which may act independently or together to cause a complex trait, genes and environmental factors may act independently or together to cause a complex trait. The actions can be additive, multiplicative, or epistatic.

One example of a multiplicative effect occurs in lung disease. Some people have a variation in the *alpha-antitrypsin* gene that moderately increases the chance that they will develop lung disease. However, if individuals with this variation also smoke regularly, they are at a greatly increased risk of getting lung disease, more than just smoking alone or just having the variation alone. Other cancer-related genes include tumor suppressor genes, whose particular alleles can influence the risk of developing breast cancer, for instance.

An Example of a Complex Trait

Alzheimer's disease is characterized by a gradual loss of brain function, usually starting with increasing loss of memory. Four different genes have been identified that can play a role in causing Alzheimer's disease. Rare mutations in each of three different genes (on different chromosomes) can each cause Alzheimer's disease. A common variation in a fourth gene, called *ApoE*, which occurs in approximately 35 percent of the population, increases the risk of developing Alzheimer's disease about threefold if a person has one copy of the allele (called *e4*), and about tenfold if they have two copies of the allele.

Taken together, these four genes account for less than half of all the genetic effects in Alzheimer's disease, indicating that additional genes that have not yet been identified are also important. In addition, environmental risk factors can have an effect. For example, taking certain anti-inflammatory medications, such as ibuprofen, reduces the risk of developing Alzheimer's disease, whereas severe head injury increases it. **SEE ALSO** ALZHEIMER'S DISEASE; BREAST CANCER; CARDIOVASCULAR DISEASE; DIABETES; EPISTASIS; INHERITANCE PATTERNS.

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Computational Biologist

A computational biologist is a scientist who develops and utilizes computational tools to analyze biological data. The Human Genome Project and other large sequencing projects have generated an extraordinary amount of data. Biologists are now faced with the challenge of extracting meaning from linear sequences composed of billions of **base pairs**. The work of computational biologists is indispensable for this task and for many other biological problems that lend themselves to computational solutions.

A basic knowledge of molecular biology and genetics is important, enabling the computational biologist to understand the issues, to interpret the meaning of results, and to communicate with research biologists who work at the laboratory bench. The actual job duties, apart from proficiency in oral and written communication and reading the literature, are very much centered around working with computers. It is, therefore, paramount that a computational biologist be highly skilled in computer technology, with expertise in hardware, software, and the principles of programming. Daily tasks range from accessing public databases and using publicly available and commercial tools for analysis, to developing novel methods for solving problems.

Until recently, there were no formal educational opportunities in computational biology at the graduate level. Therefore, most of the current practitioners and authorities in the field have a combination of degrees at the graduate (master's or doctorate) and undergraduate (baccalaureate) levels in computer science and biology. Still others hold degrees in one of the two fields and are self-taught in the discipline for which they lack formal training. Most often such individuals will have an advanced degree in computer science and will be self-taught in biology, although the converse can also be true. Nonetheless, the ability to program in a high-level language such as C++ or PERL is a major qualification.

With the release of the working draft of the human genome in 2001, computational biology has come of age. Highly qualified individuals are in demand at academic, private, and government research institutes alike. Academic institutions have taken notice and have begun implementing certificate and

base pairs two nucleotides (either DNA or RNA) linked by weak bonds



wet lab laboratory devoted to experiments using solutions, cell cultures, and other “wet” substances

graduate programs in computational biology and bioinformatics. Many more programs are in the planning stages. Thus, it will probably be desirable for future practitioners to take advantage of these specialized formal training initiatives if they wish to remain competitive.

Working environments can be fairly diverse, but be it in an office, a cubicle, a computer lab, or a corner in a **wet lab**, it invariably will consist of a desk and a secure networked UNIX workstation. Depending on the structure of the organization, one may work as part of a team or independently. It is common to have frequent contact with both biologists and programmers. The computational biologist often acts as a liaison between the two. Job opportunities range from entry level to team leader to project manager or principle investigator.

Helping biologists extract meaning from their findings is extremely rewarding. Interesting findings should be published to help research move forward. As with most scientific fields, a computational biologist enjoys the opportunity to travel to international meetings. The salary ranges from approximately \$45,000 to approximately \$150,000, depending on experience and educational attainment, and on whether one chooses to work in the public or private sector. **SEE ALSO** BIOINFORMATICS; INFORMATION SYSTEMS MANAGER; INTERNET; STATISTICAL GENETICIST.

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Conjugation

Conjugation is one of several mechanisms that bacteria use to transfer DNA, and hence new genetic information, between two cells. The other primary mechanisms are transformation, in which free DNA is transported across the cell membrane, and transduction, in which DNA is carried into the recipient cell by a bacterial virus.

The Role of Plasmids

Conjugation is about as close as single cells come to engaging in sex, and some of the terminology used to describe the process reflects that similarity. Conjugation, or mating, is a process of genetic transfer that requires cell-to-cell contact. The genetic instructions for conjugation are encoded on a double-stranded, circular piece of DNA. The circular DNA exists in the bacterial cell entirely separate from the much larger bacterial chromosome. Scientists refer to this specialized, extrachromosomal piece of DNA as a conjugative plasmid or a “fertility factor.” Cells that possess it are donor or “male” cells, and those that lack a conjugative plasmid are recipient or “female” cells.

conjugation a type of DNA exchange between bacteria



Conjugating *Escherichia coli* bacteria transfer DNA through cell-to-cell contact, which is made possible by the thread-like pilus that attaches to and reels in other cells.

There are multiple genes involved in the process of conjugation. Some of the genes code for a surface structure found on donor cells, the sex pilus. This is a threadlike tube made of protein. The sex pilus recognizes a specific attachment site on a recipient cell. When the donor cell comes near a recipient, the sex pilus attaches to the specific site and begins to retract, pulling the two cells together. This is a bit like throwing out a fishing line, hooking a fish, and pulling it into shore. The fishing analogy ends here, however. As the two cells draw close, their connection stabilizes and their outer membranes fuse together to allow the transfer of DNA from one cell to the other.

Only one of the two strands of DNA making up the plasmid passes through the fused membranes into the recipient cell. Thus DNA synthesis must occur in both donor and recipient to replace the missing strand in each. The genes encoding the enzymes responsible for this part of the conjugative process are also found on the plasmid. Once passage and synthesis are successfully completed, both donor and recipient cells contain a whole double-stranded, circular, conjugative plasmid. Thus there are now two donor cells when before there was only one. This process is so efficient that it can quickly change an entire population to donor cells. Some types of conjugative plasmids are transferred only between cells of the same species. Other types can be transferred across species; scientists call them promiscuous plasmids.

Large-Scale Gene Transfer

One of the two scientists who first described conjugation, Joshua Lederberg, ultimately won the Nobel Prize in medicine in 1958 for his discoveries concerning the organization of genetic material in bacteria. In 1946



Conjugation

"Rolling replication" transfers the F (fertility) plasmid between bacteria. The F plasmid contains the gene for formation of pili, the hairlike projection through which the plasmid passes. Adapted from Curtis, 1989.

